

JOEL HILMERSSON

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About

Versatile computational designer and software developer with a background in architecture and engineering, currently working within the field of software for design and manufacturing. My main interest lies in computational tools for geometry modeling, fabrication and design space exploration, and how we can leverage technology to turn innovative design into physical reality.

Professional Experience

Aibuild | London, UK

Early Series-A hire; core member of a three person computational geometry team responsible for extending and maintaining a slicing engine for multi-axis robotic additive manufacturing. (C++/Java)

Senior Geometry Engineer | May 2024 – Present

- Led development of geometry driven product features (e.g metal repair from 3d scans and parametric design tools) in close collaboration with R&D and platform teams
- Built a B-Rep modeling API for OpenCASCADE via SWIG and JNI, enabling parametric CAD operations and expanding the platform's design capabilities
- Developed implicit density models for print-deposition analysis, to detect defects such as porosity and overfill in the toolpathing stage

Computational Geometry Engineer | June 2023 – April 2024

- Implemented adaptive, field and topology driven, toolpath generation algorithms to automate manufacturing workflows for construction, automotive, and aerospace clients
- Developed core geometry functionality for the slicing engine from papers and technical resources (incl. UV mapping, implicit geometry, graph & topology algorithms, and surface mesh processing)

Key Skills: Java, C++, Computational Geometry, Additive Manufacturing, R&D

Generative Engineering | London, UK

Start-up company developing a platform for generative design in engineering. I joined the company in pre-seed.

Computational Engineer | Jan 2023 – April 2023

- Explored the application of generative models for design applications
- PoC integration of cloud-based FE simulations via SimScale's API (Python)

Key Skills: Python, Generative Design, Multi-Objective Optimisation

AKT II | Applied Research | London, UK

Worked as part of the software development team, which is 5 person a subgroup of the wider interdisciplinary applied research team.

Computational Designer | Feb 2020 – Jan 2023

- Lead the development of Reakt, an in-house interoperability toolkit for parametric workflows, connecting modeling and analysis software through a bespoke object model (C#)
- Supported projects with structural analysis (FEA) and parametric modeling, collaborating with architects such as Foster + Partners and Zaha Hadid
- Conducted internal R&D on meshing for FEA, workflow automation, and computational modeling, providing custom tools for project teams (C#, C++)
- Contributed to the Horizon 2020 PrismArch project as a domain expert

Key Skills: Rhino, Grasshopper, NURBS, C#, Python, FEA Software (+APIs)

Internships & Part time work

• Sunneroe Architects | Gothenburg, Sweden | 2019

Part-time external consultant. Developed a generative grasshopper toolkit for automated apartment layout design. Using Rhino, Grasshopper, Python

• Bollinger + Grohmann | Oslo, Norway | 2018

FEA Analysis & Multi-Objective Optimization.
Using Rhino, Grasshopper, C#, Karamba, RFem

• Knippers Helbig | Stuttgart, Germany | 2015 - 2016

Computational design for complex geometry and FEA for projects globally. Using Rhino, Grasshopper, C#, Sofistik

Education

Chalmers University of Technology | Gothenburg, Sweden

M.Arch | Architecture & Urban Design | Sept 2017 - Aug 2019

- Master in architecture & urban design with a focus on digital design and robotic fabrication

M.Sc | Structural Engineering & Building Technology | Sept 2016 – Aug 2019

- Master in engineering with a focus on structural mechanics and FEA.
- Thesis: Isogeometric analysis & form finding - Wrote a design tool for bending active thin shells using nonlinear NURBS based finite element methods (Grasshopper, MATLAB, C#)

B.Sc | Architecture & Engineering | Aug 2012 – June 2015

- Interdisciplinary bachelor's degree, combining design studies with engineering and maths

Delft University of Technology | Delft, Netherlands

Erasmus | Computational Mechanics | Sept 2016 – Aug 2017

- 1 Year ERASMUS scholarship as part of engineering master. Studies in engineering specializing in computational mechanics.

Academic & Research Projects

PrismArch | Horizon2020 | 2020 – 2023

An interdisciplinary research project bridging academia and industry to explore VR-based design workflows. Collaborators included Zaha Hadid Architects, ETH Zurich, and the University of Malta. Funded by the EU's Horizon 2020 program

- Applied research on evolutionary algorithms in collaboration with AI researchers at the University of Malta, resulting in a shape optimization framework for structural design (C#)

- Authored key sections of Horizon 2020 deliverables, on the topics of cross-disciplinary AEC ontologies and algorithmic design principles

Conference Papers

• IASS 2023 | Design Space Exploration of Shell Structures Using Quality Diversity Algorithms

K. Sfikas, A. Liapis, [J. Hilmersson](#), J. Dudley, E. Tibuzzi, G. Yannakakis

• IASS 2021 | The Geldeford Riband

D. Godfrey, J. Dudley, [J. Hilmersson](#)

• IASS 2019 | Isogeometric Analysis and Form Finding of Thin Elastic Shells

[J. Hilmersson](#), J. Olsson, M. Ander, F. Larsson

Technical Skills

Programming languages: C#, Java, Python, C++ (17/20), Rust

Geometry APIs: RhinoCommon, OpenCASCADE, CGAL

Numerical/Computing: Eigen, NumPy, Jax (familiar), MATLAB

Developer Tools: JNI, cmake, git, swig

Simulation & FEA: Sofistik, SAP2000, ETABS, RFem, Karamba3d (API)

3D modeling: Rhino, Grasshopper, Blender

Domain Expertise:

computational geometry (*B-Rep, implicit*), parametric modeling, finite element analysis, design space exploration, optimisation, additive manufacturing